

Minh Hai NGUYEN

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I am a PhD student in MAMBO team at Université de Toulouse (expected graduation: December 2026), advised by Prof. Pierre Weiss and Prof. Edouard Pauwels. My research focuses on generative models for inverse problems in imaging, with applications to microscopy.

EDUCATION

PhD in Applied Mathematics

[Toulouse University, France](#)

Advised by Prof. Pierre Weiss and Prof. Edouard Pauwels

Research topics: Inverse Problems in Imaging, applications to Microscopy

2023 -

Master of Engineering in Applied Mathematics

[INSA Toulouse, France](#)

Major in AI / Data Science

(*Location: 2018-2020 in Vietnam and 2020-2023 in France)

2018 - 2023

EXPERIENCE

Research Engineer

[Agenium Space](#)

Correcting micro-vibrations in satellite imaging with pushbroom cameras.

Dec. 2022 - Sept. 2023

Research Intern

[Institut Mathématiques de Toulouse](#)

Product-convolution neural networks: applications to image deblurring with space-varying blurs.

Jun. 2022 - Sept. 2022

PUBLICATIONS

- [1] M. H. Nguyen, E. Pauwels, and P. Weiss. "How Diffusion Prior Landscapes Shape the Posterior in Blind Deconvolution". In: *Pre-print* (2024).
- [2] M. H. Nguyen, F. de Vieilleville, and P. Weiss. "DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras". In: *IEEE Transactions on Geoscience and Remote Sensing* 62 (2024), pp. 1–9.
- [3] M. H. Nguyen and P. Weiss. "Comparing Plug-and-Play and Unrolled networks". In: *Pre-print* (2024).

LANGUAGE

English: fluent

French: fluent

Vietnamese: native

PROGRAMMING LANGUAGE

Python (advanced): PyTorch (DDP, AMP)

C/C++ (intermediate)

OTHER SERVICES

Open source project

DeepInverse

[PyTorch library for solving imaging inverse problems using deep learning](#)

[Github Repository](#)

Teaching

- Signal Processing (1st Master level) at INSA Toulouse, in 2024.

- Stochastic Optimization and Learning (1st Master level) at INSA Toulouse / ENSEEIHT, in 2024.

HONOR AND AWARDS

French National Prize for Open Science Library for DeeplInverse.

Documentation prize

By Ministère de l'Enseignement Supérieur et de la Recherche

Nov. 2024

IA Pau Competition

Best project award

By IA Pau organization, funded by Boavizta (1500eur prize).

2023